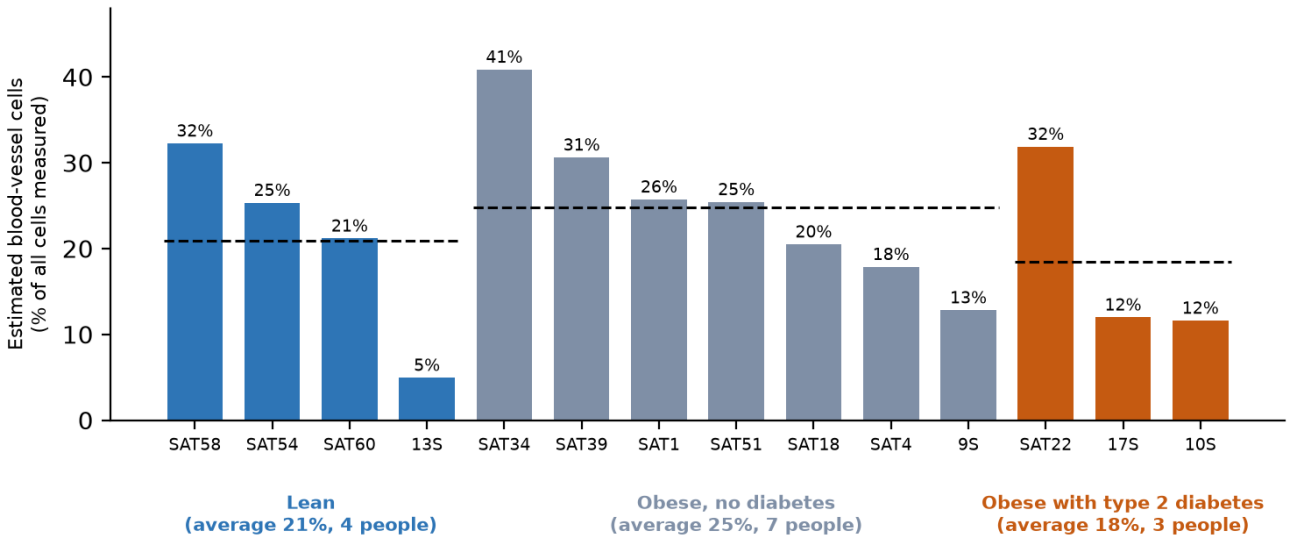
Blood-vessel cells in human fat: lean, obese, and obese with type 2 diabetes

Public dataset GSE268904: fat tissue under the skin from 14 people. Counts below are estimates from gene activity, not a laboratory identification.

131,641	14	24,070	18%
individual cells (nuclei) measured	people (samples)	estimated blood-vessel cells	of all cells



Group	People	Cells measured	Est. blood-vessel cells	Average per person	Range across people
Lean	4	37,418	5,762	21%	5% to 32%
Obese, no diabetes	7	58,141	13,211	25%	13% to 41%
Obese with type 2 diabetes	3	36,082	5,097	18%	12% to 32%

What this shows

About 1 in 5 cells measured looks like a blood-vessel lining cell. The share is very different from person to person (from 5% to 41%), and the groups overlap heavily.

Please read with care

- Only 3 people have type 2 diabetes, so group differences are exploratory and prove nothing yet.
- The obese groups are much heavier (BMI about 50) than the lean group (about 23), so any difference may reflect weight, not diabetes.
- Cells are not people: the real sample size is 14.

How a cell was counted as a blood-vessel cell (endo_specific)

A cell is counted when (1) at least 3 of 6 blood-vessel marker genes (PECAM1, VWF, CLDN5, EGFL7, FLT1, PTPRB) are active in it, (2) blood-vessel genes are its strongest signal compared with fat, connective-tissue, immune and vessel-wall genes, and (3) that signal is at least twice as strong as any of the others. This guards against cells that only look blood-vessel-like because of stray RNA in the sample. It is still a marker-based estimate, not a formal cell-type annotation.

Source: GEO GSE268904 (Aarhus University), analysed with scripts/GSE268904_extract.R. Counts: barcodes_clean table, 131,641 nuclei.